Azure

- Azure, also known as Microsoft Azure, is a cloud computing platform and set of services provided by Microsoft. It offers a wide range of cloud services, including computing, storage, networking, databases, analytics, artificial intelligence, machine learning, Internet of Things (IoT), and more.
- Azure enables organizations to build, deploy, and manage applications and services through Microsoft's global network of data centers. It supports various programming languages, frameworks, and tools, providing flexibility for developers and IT professionals.
- Key features and components of Azure include:
    1. Virtual Machines (VMs): Allows users to deploy and manage virtual machines running Windows or Linux in the cloud.
    2. Azure App Service: Provides a platform for building, deploying, and scaling web applications and APIs.
    3. Azure Storage: Offers scalable, secure, and highly available storage solutions, including blob storage, file storage, and disk storage.
    4. Azure SQL Database: A fully managed relational database service based on Microsoft SQL Server.
    5. Azure Kubernetes Service (AKS): Simplifies the deployment, management, and scaling of containerized applications using Kubernetes.
    6. Azure Active Directory (AD): Provides identity and access management services, including single sign-on, multi-factor authentication, and role-based access control.
    7. Azure Functions: Enables serverless computing, allowing developers to run code in response to events without managing infrastructure.
    8. Azure Cognitive Services: Offers a set of APIs and services for adding artificial intelligence and machine learning capabilities to applications, including vision, speech, language, and decision services.
    9. Azure IoT Hub: Provides a managed service for connecting, monitoring, and managing IoT devices at scale.
    10. Azure DevOps: Offers a suite of tools for agile planning, version control, continuous integration, continuous delivery, and monitoring of applications and services.
- Benefits:
    - Scalability: Easily scale resources up or down based on demand.
    - Flexibility: Supports various programming languages, frameworks, and tools.
    - Security: Provides robust security and compliance features.
    - Integration: Integrates with existing on-premises infrastructure and applications.
    - AI and ML Capabilities: Offers artificial intelligence and machine learning services.

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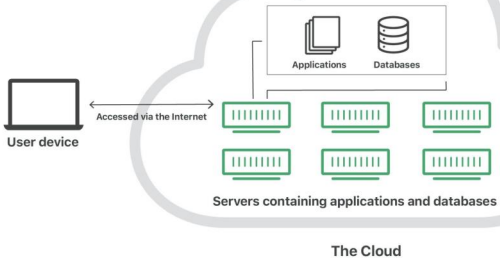
Cloud

- Cloud computing refers to the delivery of computing services, including servers, storage, databases, networking, software, and analytics, over the internet ("the cloud") to offer faster innovation, flexible resources, and economies of scale
- Cloud computing allows users and companies to access and use computing resources on-demand, eliminating the need for managing physical servers or running software applications on their own machines.
- Cloud Providers -
    1. Amazon Web Services (AWS)
    2. Microsoft Azure
    3. Google Cloud Platform (GCP)
    4. Alibaba Cloud:
    5. IBM Cloud



How does cloud computing work?

Cloud computing works by utilizing remote servers hosted on the internet to store, manage, and process data, as opposed to relying on local servers or personal computers. The process involves several key components and steps:

1. Infrastructure:
   Cloud service providers maintain vast data centers with powerful servers, storage systems, networking equipment, and other hardware infrastructure. These data centers are located in various geographic regions around the world.

2. Virtualization:
   Virtualization technology enables the creation of virtual instances of servers, storage, and other resources. This allows multiple users to share the same physical hardware while maintaining isolation and security for their data and applications.

3. Service Models:
   Cloud computing offers different service models:
   - Infrastructure as a Service (IaaS): Provides virtualized computing resources over the internet, such as virtual machines, storage, and networking.
   - Platform as a Service (PaaS): Offers a platform allowing developers to build, deploy, and manage applications without worrying about the underlying infrastructure.
   - Software as a Service (SaaS): Delivers software applications over the internet on a subscription basis, eliminating the need for users to install, maintain, and update the software locally.

4. Access and Delivery:
Users access cloud services and applications through the internet using web browsers, dedicated client applications, or APIs (Application Programming Interfaces). Data and applications hosted in the cloud can be accessed from any device with an internet connection.

5. Scalability and Elasticity:
Cloud computing platforms offer scalability, allowing users to easily scale up or down their resources based on demand. This ensures that organizations can handle fluctuations in workload without over-provisioning or under-provisioning resources.

6. Security and Compliance:
Cloud service providers implement robust security measures to protect data and infrastructure, including encryption, access controls, network security, and compliance certifications. Users are also responsible for implementing appropriate security measures to protect their data in the cloud.

Types of Cloud Service

1. Software-as-a-Service (SaaS):
    a. Applications hosted on cloud servers, accessed over the internet.
    b. Users don't need to install software locally; they access it through a web browser.
    c. Examples: Salesforce, MailChimp, Slack.
    d. Analogy: Renting a fully furnished house where the landlord maintains the property, and the tenant uses it as if they own it.
2. Platform-as-a-Service (PaaS):
    a. Provides tools and infrastructure for building applications over the internet.
    b. Companies pay for development tools, infrastructure, and operating systems.
    c. Examples: Heroku, Microsoft Azure.
    d. Analogy: Renting all the tools and equipment necessary for building a house instead of renting the house itself.
3. Infrastructure-as-a-Service (IaaS):
    a. Companies rent servers and storage from a cloud provider.
    b. Use cloud infrastructure to build and manage applications.
    c. Examples: DigitalOcean, Google Compute Engine.
    d. Analogy: Leasing a plot of land to build whatever you want, but you need to provide your own building equipment and materials.
4. Function-as-a-Service (FaaS):
    a. Breaks cloud applications into smaller components that run only when needed.
    b. Also known as serverless computing.
    c. Functions scale up or duplicate automatically as needed.
    d. Examples: AWS Lambda, Google Cloud Functions.
    e. Analogy: Renting a house bit by bit, paying only for the parts you use at any given time, with rooms expanding on demand.

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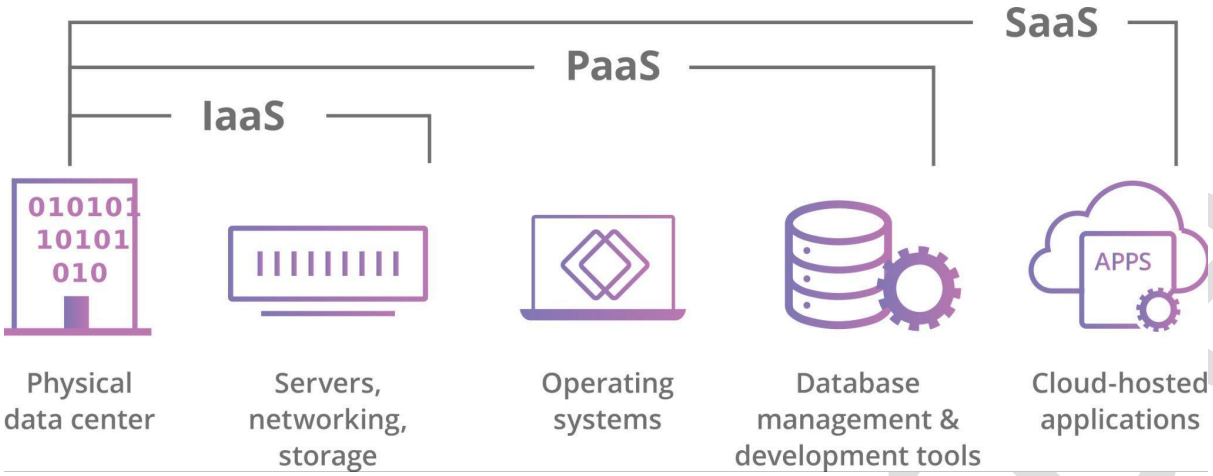

What are the different types of cloud deployments?

There are two main categories of cloud deployments, differentiated by ownership and location: public clouds and private clouds. Additionally, there are hybrid and multi-cloud models that combine elements of both. Here's a breakdown of each type:

1. Public Cloud:
    - Ownership: Owned and operated by a cloud service provider (CSP) like Amazon Web Services (AWS), Microsoft Azure, or Google Cloud Platform (GCP).
    - Location: Public cloud infrastructure is spread across various geographical locations, offering scalability and redundancy.
    - Access: Public clouds are accessible to the general public over the internet.
    - Cost: Typically pay-as-you-go pricing model, making it cost-effective for variable workloads.
    - Security: Providers implement robust security measures, but users are ultimately responsible for securing their data and applications within the cloud environment.
    - Examples: AWS, Azure, GCP, Alibaba Cloud, IBM Cloud.
2. Private Cloud:
    - Ownership: Owned and operated by a single organization, either on-premises or hosted by a service provider in a dedicated environment.
    - Location: Private cloud infrastructure is located within the organization's data center or a dedicated facility.
    - Access: Restricted to authorized users within the organization.
    - Cost: Can be more expensive than public cloud due to the upfront investment in infrastructure and ongoing maintenance.
    - Security: Offers greater control over data security and privacy as the organization manages the infrastructure.

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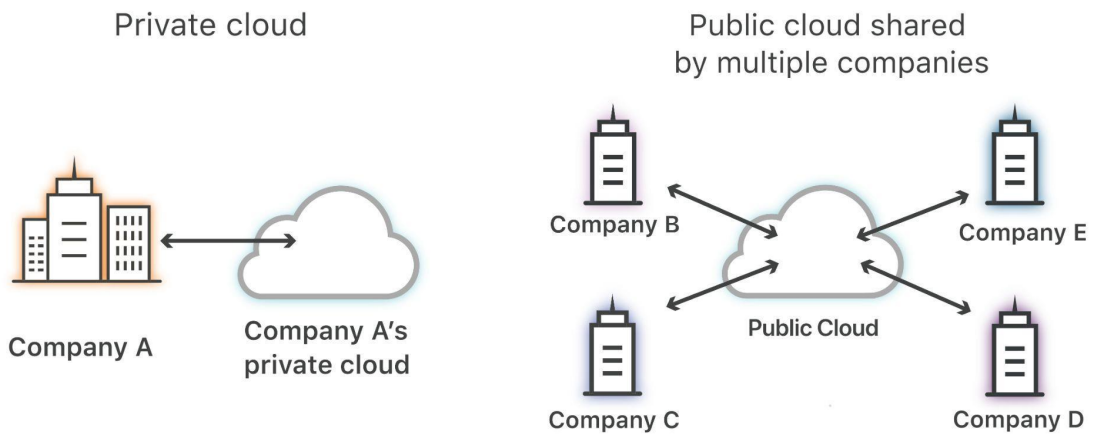
- Examples: Can be implemented using various technologies and vendors, depending on the organization's needs



3. Hybrid Cloud:
    - Combination: Combines elements of both public and private clouds, allowing organizations to leverage the benefits of each.
    - Functionality: Enables organizations to move workloads between public and private clouds based on factors like security, cost, and performance requirements.
    - Benefits: Offers flexibility, scalability, and control over data and applications.
    - Complexity: Requires careful planning and management to ensure seamless integration and security between the different cloud environments.
4. Multi-Cloud:
    - Multiple Clouds: Utilizes multiple public clouds or a combination of public and private clouds from different providers.
    - Benefits: Enhances flexibility, redundancy, and potential cost savings by allowing organizations to choose the best cloud provider for specific workloads based on factors like pricing, features, and performance.
    - Complexity: Managing multiple cloud environments from different providers can be challenging and requires careful planning and expertise.

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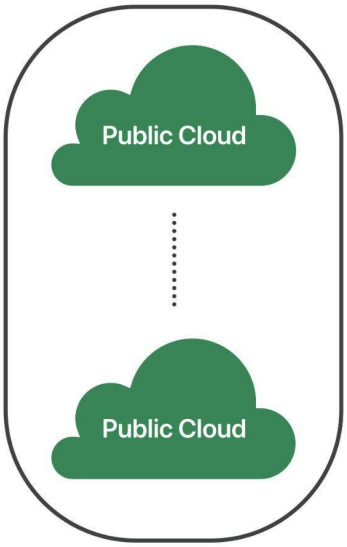
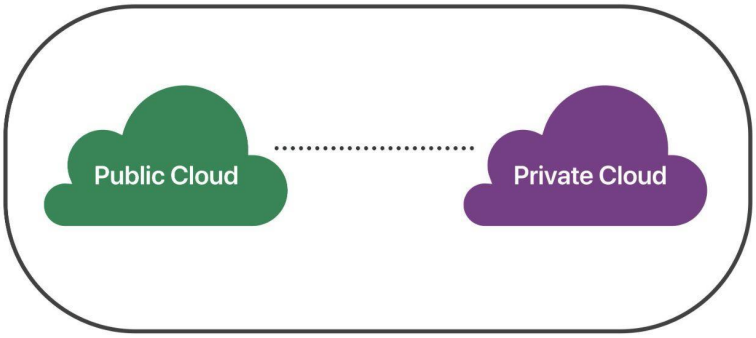
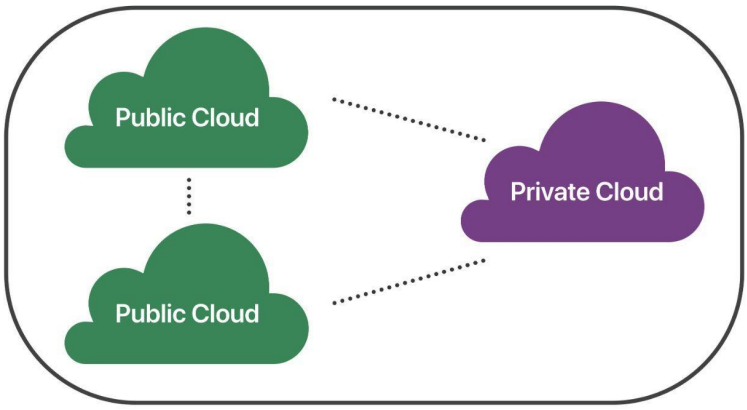


Multicloud



Hybrid Cloud



Multi/Hybrid Cloud

ETL (Extract, Transform, Load)

Extract

- Definition: Extraction involves retrieving data from various data sources. These sources can be heterogeneous and might include databases, flat files, APIs, and other structured or unstructured data stores.
- Key Activities:

    - Identifying Data Sources: Determining the data sources that need to be integrated.
    - Data Extraction: Extracting data using methods like querying databases, reading files, calling APIs, or scraping web data.
    - Handling Source Data Formats: Managing different data formats and ensuring compatibility with downstream processes.
- Challenges:
    - Heterogeneity: Dealing with different data formats, structures, and types.
    - Volume: Extracting large volumes of data efficiently.
    - Latency: Minimizing the time taken to extract data, especially in real-time or near-real-time systems.
- Tools:
    - SQL queries, Python scripts, ETL tools like Apache Nifi, Talend, Informatica.

Transform

- Definition: Transformation involves cleaning, enriching, and modifying the extracted data to fit the desired format and structure of the target data warehouse. This step ensures data quality and consistency.
- Key Activities:
    - Data Cleaning: Removing duplicates, correcting errors, handling missing values.
    - Data Enrichment: Adding context to the data, such as geocoding addresses or deriving new attributes.
    - Data Aggregation: Summarizing data, such as calculating totals or averages.
    - Data Normalization: Ensuring data conforms to a standard format or scale.
    - Data Validation: Ensuring data meets the required quality and business rules.
- Challenges:
    - Complexity: Managing complex transformation logic and business rules.
    - Performance: Efficiently transforming large datasets.
    - Data Quality: Ensuring transformed data is accurate and reliable.
- Tools:
    - ETL tools like Informatica, Talend, Apache Nifi; scripting languages like Python or R; SQL.

Load

- Definition: Loading involves transferring the transformed data into the target data warehouse or data repository. This step can be executed in various modes, depending on the requirements, such as batch loading, incremental loading, or real-time loading.
- Key Activities:
    - Data Insertion: Inserting data into the target system using bulk load or incremental updates.
    - Indexing: Creating indexes to improve query performance.
    - Partitioning: Splitting data into partitions for efficient access and management.
    - Verification: Ensuring data integrity and accuracy after loading.

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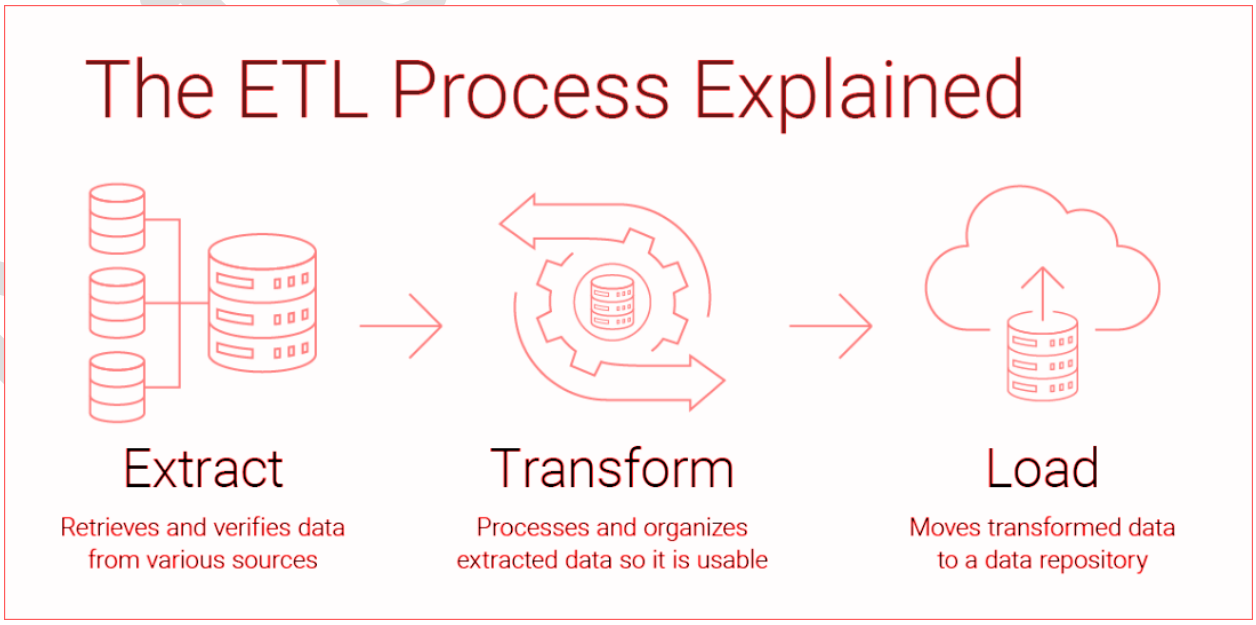
- Challenges:
    - Volume: Loading large volumes of data efficiently.
    - Consistency: Ensuring data consistency and integrity in the target system.
    - Performance: Minimizing the impact on the target system's performance during loading.
- Tools:
    - Database management systems (DBMS) like MySQL, PostgreSQL, Oracle; ETL tools like Informatica, Talend.
    - ETL Process Flow
    - Data Extraction: Raw data is extracted from multiple heterogeneous sources.
    - Data Transformation: Extracted data is cleaned, enriched, and transformed to meet the requirements of the target schema.
    - Data Loading: Transformed data is loaded into the data warehouse or other target repository.

- Benefits of ETL:
    - ETL enables organizations to integrate and analyze data from disparate sources, providing valuable insights for decision-making and business intelligence.
    - It helps ensure data quality, consistency, and accessibility by cleaning, transforming, and organizing data before loading it into target systems.
    - ETL improves data governance and compliance by enforcing data quality standards and security measures throughout the data integration process
- Challenges:
    - ETL processes can be complex and time-consuming, especially when dealing with large volumes of data and multiple data sources.
    - Maintaining data quality and consistency throughout the ETL pipeline is crucial.Ensuring scalability and performance to handle increasing data volumes and processing requirements.



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Types of Data

Feature	Structured Data	Semi-structured Data	Unstructured Data
Definition	Highly organized data with a predefined format (schema)	Data with some internal organization, often using tags or markers, but lacking a strict schema	Data that lacks a predefined format or schema
Examples	Relational database tables, CSV files.	JSON, XML files, log files, HTML	Text documents, emails, social media posts, images, videos
Storage	Relational databases, data warehouses	File systems, NoSQL databases	File systems, specialized data lakes
Analysis	Easy to search, filter, and analyze due to its structured nature	Requires parsing and interpretation before analysis, often using specific tools	Requires significant processing and transformation before analysis, often using machine learning and natural language processing
Flexibility	Less flexible, requires schema modification for changes	More flexible than structured data, can accommodate some variations	Most flexible, easily adapts to different data formats
Scalability	Generally scales well with traditional database technologies	Can scale well with specialized NoSQL databases	Can be challenging to scale due to lack of structure
Access	Accessible using SQL or structured queries.	Accessible using semi-structured data processing tools.	Accessible using text mining or natural language processing techniques.

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Resource Group

- Definition: A Resource Group is a container that holds related resources for an Azure solution. These resources could be anything from virtual machines, storage accounts, web apps, databases, and more.
- Purpose: It helps manage and organize resources in a way that simplifies their lifecycle (such as deployment, monitoring, and billing).
- Key Features
    - Lifecycle Management: Resources within a Resource Group can be managed as a single entity. For example, you can delete the entire Resource Group, which deletes all the resources contained within it.
    - Role-Based Access Control (RBAC): Resource Groups allow you to assign specific permissions to users or groups for managing resources. This way, you can control who has access to what.
    - Cost Management: Resource Groups help in tracking and managing costs by grouping resources with similar cost centers or projects.
    - Deployment: Using Resource Groups, you can deploy, update, and delete all the resources in a single, coordinated operation.
    - Tagging: You can apply tags to Resource Groups and the resources within them for better organization, reporting, and management.
- Best Practices
    - Logical Grouping: Group resources that share the same lifecycle. For instance, all resources for a web application might go into one Resource Group.
    - Access Management: Use RBAC to control access to resources based on roles and responsibilities.
    - Naming Conventions: Use consistent naming conventions to easily identify the purpose of Resource Groups and the resources within them.
    - Tagging Strategy: Implement a tagging strategy to better manage and report on resources.

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Azure Storage

- The Azure Storage platform is a highly scalable and durable cloud storage solution designed to store large amounts of unstructured and structured data.
- It provides a variety of storage services to cater to different use cases, including object storage, file storage, and more.
- Types of Azure Storage platform:
    1. Azure Blob Storage
        - Purpose: Store large amounts of unstructured data, such as text or binary data.
        - Types of Blobs:
            - ➢ Block Blobs: Ideal for storing text and binary data, such as documents and media files.
            - ➢ Append Blobs: Optimized for append operations, commonly used for logging.
            - ➢ Page Blobs: Designed for frequent read/write operations, used for virtual hard disk files (VHDs) and other random-access storage needs.
        - Use Cases: Backup and restore, disaster recovery, archival storage, and serving images or documents directly to a browser.
    2. Azure Data Lake Storage (ADLS) Gen2
        - Purpose: Combines the capabilities of Azure Blob Storage with a file system for big data analytics.
        - Features: Hierarchical namespace, scalable storage, high throughput, and integration with big data frameworks like Hadoop and Spark.
        - Use Cases: Big data analytics, data warehousing, machine learning, and data lake implementations.
    3. Azure Files
        - Purpose: Fully managed file shares in the cloud accessible via the Server Message Block (SMB) protocol.
        - Features: Shared access, REST API access, integration with on-premises systems using Azure File Sync.
        - Use Cases: Lift and shift applications, file server replacement, shared file access across multiple virtual machines.
    4. Azure Queue Storage
        - Purpose: Message queue service for storing large numbers of messages that can be accessed from anywhere in the world.
        - Features: Simple and cost-effective messaging for workflows, decoupling components of cloud applications.
        - Use Cases: Asynchronous task processing, workload distribution, and communication between cloud services.
    5. Azure Table Storage
        - Purpose: NoSQL key-value store for storing structured, non-relational data.
        - Features: High availability, scalability, and low-latency access.

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- Use Cases: Storing structured data such as user profiles, product information, and other metadata.
- Key Features of Azure Storage:
- Durability and Availability: Data is automatically replicated within the region and optionally to another region for disaster recovery.
- Security: Data encryption at rest and in transit, role-based access control (RBAC), and integration with Azure Active Directory.
- Scalability: Easily scale to accommodate growing data volumes and fluctuating workloads.
- Cost-Effectiveness: Various pricing tiers and options to optimize storage costs based on access patterns and data redundancy requirements.

Service	**Azure Blob Storage**	**Azure Data Lake Storage (ADLS) Gen2**	**Azure Files**	**Azure Queue Storage**	**Azure Table Storage**
Purpose	Store large amounts of unstructured data like text or binary data.	Big data analytics with integrated capabilities of Blob Storage and file system.	Managed file shares accessible via SMB protocol.	Store large numbers of messages for asynchronous communication.	NoSQL key-value store for structured data.
Key Features	- Block, Append, and Page Blobs.	- Hierarchical namespace.	- Shared access with REST API.	- Simple and cost-effective messaging.	- Schema-less design.
	- Backup, disaster recovery, media serving.	- Integration with big data frameworks (Hadoop, Spark).	- Integration with Azure File Sync.	- High availability and scalability.	- Low-latency access.
	- High durability and availability.	- Scalable and high throughput.	- Fully managed and scalable.	- Easily decouple cloud applications.	- High availability and scalability.
Use Cases	Backup and restore, media storage, serving documents or images.	Big data analytics, data warehousing, machine learning.	Lift and shift applications, file server replacement, shared file access.	Asynchronous task processing, workload distribution, communication between services.	Storing structured data such as user profiles, metadata, and IoT data.
Logo					

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1. Azure Blob Storage:
    - Purpose:
        - Azure Blob Storage is a scalable object storage service provided by Microsoft Azure, designed to store and manage unstructured data efficiently and securely in the cloud.
    - Key Features:
        - Blobs: Objects stored in Azure Blob Storage, optimized for storing large amounts of unstructured data such as documents, images, videos, and logs.
        - Containers: Organizes blobs into containers, similar to folders in a file system, providing a logical grouping mechanism.
        - Scalability: Automatically scales to meet the demands of any application and data size, from a few terabytes to petabytes of data.
        - Data Redundancy: Offers built-in redundancy options (locally redundant storage, geo-redundant storage) to ensure data durability and high availability.
        - Storage Tiers: Provides different access tiers (hot, cool, archive) to optimize storage costs based on data access patterns.
        - Lifecycle Management: Automates the movement and deletion of blobs between different access tiers based on defined policies.
        - Security: Data at rest is encrypted using Azure Storage Service Encryption (SSE) by default, with support for customer-managed keys (CMKs) for enhanced security compliance.
        - Integration: Seamlessly integrates with other Azure services like Azure Data Factory, Azure Functions, and Azure Databricks for data processing, analytics, and content delivery.
    - Use Cases:
        - Data Backup and Archive: Store backup copies of data or archive rarely accessed data.
        - Media Storage and Streaming: Serve multimedia files such as images, videos, and audio files for web applications.
        - Big Data Analytics: Store and analyze data for big data solutions, machine learning models, and data lakes.

2. Azure Data Lake Storage Gen2 (ADLS Gen2):
    - Purpose:
        - Azure Data Lake Storage Gen2 combines the capabilities of Azure Blob Storage and Azure Data Lake Storage Gen1 into a single service, optimized for big data analytics and large-scale data processing.
    - Key Features:
        - Hierarchical Namespace: Provides a hierarchical file system, allowing organizations to organize data into directories and subdirectories for efficient data management and access control.
        - Integration: Integrates seamlessly with Azure services like Azure Databricks, Azure Synapse Analytics, and Azure Data Factory for data processing, analytics, and ETL operations.

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        - Security: Supports Azure RBAC, ACLs, and Azure AD authentication for fine-grained access control and data security compliance.
        - Scalability: Scales to handle petabytes of data with high throughput and low latency, ideal for big data processing and analytics workloads.
        - Storage Tiers: Offers different access tiers (hot, cool, archive) to optimize storage costs based on data access patterns.
        - Data Redundancy: Provides built-in redundancy options to ensure data durability and high availability.
        - Encryption: Data at rest can be encrypted using SSE (Server-Side Encryption) or customer-managed keys (CMKs) for enhanced security.
    - Use Cases:
        - Big Data Analytics: Store and analyze large volumes of structured and unstructured data for business insights and machine learning.
        - Data Warehousing: Supports data warehousing solutions, enabling high-performance analytics and reporting.
        - Machine Learning and AI: Provides a unified data repository for training machine learning models and running AI algorithms.

3. Azure Table Storage:
    - Purpose:
        - Azure Table storage is a NoSQL key-value store designed to store structured data in the cloud. It offers schemaless storage, allowing flexibility in data modeling without predefined schemas.
    - Key Concepts:
        - Tables: Collections of entities, each containing properties stored as key-value pairs.
        - Partition Key and Row Key: Unique identifiers for entities within a table, enabling efficient querying and scalability.
        - Scalability: Automatically scales based on demand, accommodating large volumes of structured data.
    - Use Cases:
        - Store user profiles, configuration settings, and metadata.
        - Back-end storage for web applications requiring fast access to structured data.
        - Integration: Integrates with Azure services like Azure Functions and Azure Data Factory for data processing and automation.
    - Security and Compliance:
        - Implements Azure RBAC, Shared Access Signatures (SAS), and Azure AD integration for secure access control.
        - Data at rest is encrypted using Azure Storage Service Encryption (SSE) for enhanced security.

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4. Azure Queue Storage:
    - Purpose:
        - Azure Queue storage provides a reliable, scalable message queuing service for asynchronous communication between application components.
    - Key Concepts:
        - Messages and Queues: Messages are stored in queues and processed asynchronously by consumer applications.
        - At-Least-Once Delivery: Ensures reliable delivery of messages to consumers.
        - Visibility Timeout: Controls message visibility to prevent concurrent processing by multiple consumers.
    - Use Cases:
        - Task distribution and load leveling in distributed systems.
        - Communication between microservices and application components.
        - Integration: Supports SDKs, REST APIs, and Azure Functions for seamless integration with cloud applications.
    - Scalability and Performance:
        - Scales automatically to handle varying message volumes and processing loads.
        - Optimized for high-throughput scenarios with low latency.

5. Azure File Share:
    - Purpose:
        - Azure File Share provides fully managed file shares in the cloud, accessible via the Server Message Block (SMB) protocol.
    - Key Concepts:
        - Managed File Shares: Offers scalable and accessible file storage for cloud-based applications.
        - Cross-Platform Access: Supports SMB protocol for access from Windows, Linux, and macOS environments.
    - Use Cases:
        - Replace or supplement on-premises file servers with cloud-based file shares.
        - Share files across multiple Azure VMs and on-premises systems.
        - Integration: Integrates with Azure Backup, Azure AD for identity-based access control, and Azure Virtual Machines.
    - Security and Compliance:
        - Data at rest is encrypted using Azure Storage Service Encryption (SSE).
        - Access control managed through Azure RBAC, ensuring secure file access.

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6. Azure Disk Storage:
    - Purpose:
        - Azure Disk Storage provides persistent block storage for Azure Virtual Machines (VMs) and Azure Kubernetes Service (AKS) clusters.
    - Types of Disks:
        - Managed Disks: Simplify disk management with Azure managing provisioning, scaling, and availability.
        - Unmanaged Disks: Traditional approach where users manage disks and storage accounts.
    - Features and Use Cases:
        - Virtual Machines: Attach managed or unmanaged disks for operating system, application data, or temporary storage needs.
        - AKS Clusters: Use managed disks as persistent volumes for stateful applications in Kubernetes clusters.
    - Performance and Scalability:
        - Offers different performance tiers (Standard HDD, Standard SSD, Premium SSD) to meet various workload requirements.
        - Automatically scales based on demand, ensuring high availability and durability.
    - Security and Compliance:
        - Data at rest encrypted using Azure Storage Service Encryption (SSE).
        - Access control managed through Azure RBAC, ensuring secure access to disk resources.

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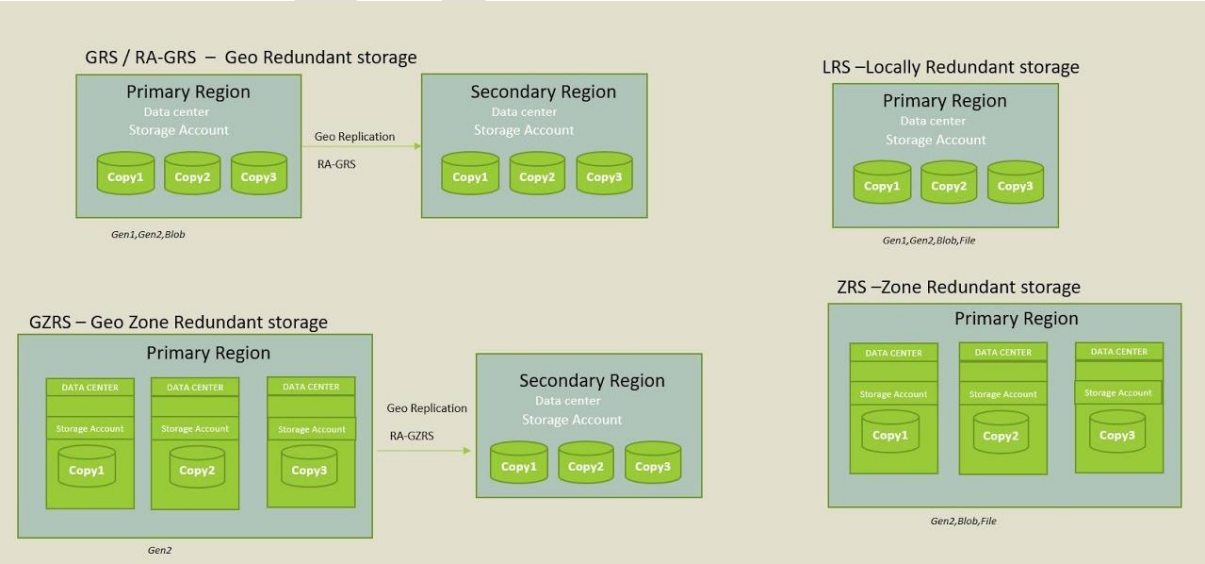
Azure Storage Redundancy

- Data replication refers to the process of automatically copying and storing data across multiple locations to ensure high availability, durability, and resilience against potential failures or disasters.

Replication Type	Description	Use Case
Locally Redundant Storage (LRS)	Copies data synchronously within a single data center to protect against local hardware failures.	Scenarios where data residency requirements are within a specific geographic region and high availability within that region is sufficient.
Zone-Redundant Storage (ZRS)	Replicates data synchronously across multiple availability zones within a region.	Applications requiring higher availability than LRS, protecting against zone-level failures within a region.
Geo-Redundant Storage (GRS)	Replicates data synchronously across paired regions, with asynchronous copies in a secondary region.	Ensures higher durability and availability, protecting against regional disasters and ensuring data access during region-wide outages.
Read-Access Geo-Redundant Storage (RA-GRS)	Extends GRS by providing read access to data replicated in the secondary region.	Applications needing read access to data in the secondary region for redundancy or disaster recovery purposes.



- Key Considerations for Data Replication in Azure:
    - Durability: Ensures data integrity and availability through redundant copies stored across multiple locations.
    - Availability: Provides continuous access to data even in the event of localized failures or disasters.
    - Performance: Minimizes latency by synchronizing data replication efficiently across Azure regions.
    - Cost: Different replication options may impact cost, with higher redundancy levels typically associated with higher costs.

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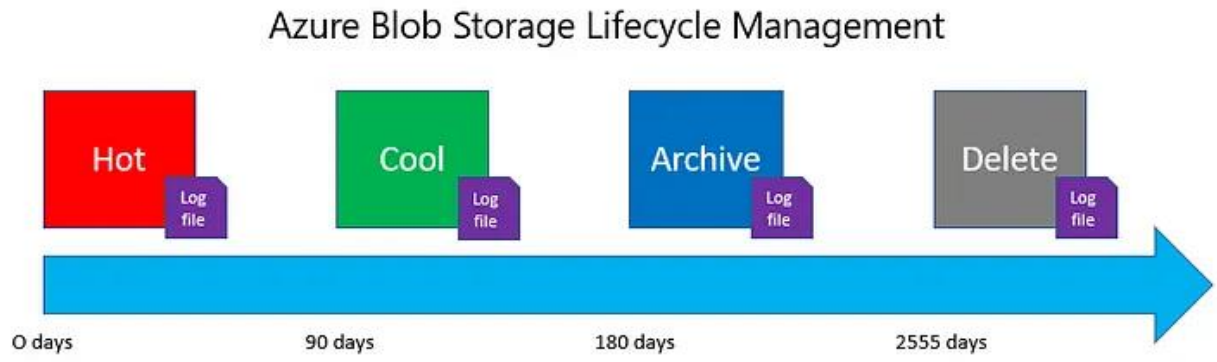
Blob Access Tier

- In Azure Blob Storage, the access tier determines the availability and pricing model for storing blob data.

Access Tier	Description	Use Cases	Benefits	Cost Considerations
Hot Access Tier	Optimized for frequent access and high-performance scenarios.	- Active workloads. - Streaming data. - Business-critical applications.	- High availability. - Low access latency.	Higher storage costs but lower access and data retrieval costs compared to other tiers.
Cool Access Tier	Optimized for infrequently accessed data stored for at least 30 days.	- Backup. - Disaster recovery. - Long-term storage with occasional access.	- Lower storage costs compared to Hot tier .- Quick retrieval for infrequent access.	Reduced storage costs but slightly higher access and data retrieval costs compared to Hot tier.
Cold Access Tier	Optimized for data that is rarely accessed and stored for at least 180 days.	- Long-term backup.-Regulatory archives. - Retaining data with minimal access requirements.	- Lower storage costs compared to Cool tier. - Suitable for data with long-term retention needs.	Lower storage costs but higher costs for data retrieval and access compared to Cool tier.
Archive Access Tier	Designed for data rarely accessed and stored for long-term retention.	- Compliance. - Legal hold. - Regulatory requirements where data is rarely accessed.	- Lowest storage costs among all tiers. - Suitable for data offline for extended periods.	Significantly lower storage costs but higher costs for data retrieval and access compared to other tiers.



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Secure Storage Account

- Securing Azure Storage accounts, including Blob Storage, is crucial for protecting data integrity, confidentiality, and availability.

Security Aspect	Description	Best Practices
Authentication and Authorization	Manage access to Azure Storage resources using Azure Active Directory (Azure AD) integration and Shared Access Signatures (SAS).	- Use Azure AD for centralized access management. - Generate limited-time SAS tokens with specific permissions. - Implement RBAC for fine-grained access control.
Encryption	Encrypt data at rest and in transit to protect confidentiality and integrity.	- Enable Azure Storage Service Encryption (SSE) for data at rest. - Use HTTPS and secure protocols for data in transit.
Network Security	Restrict access to storage accounts and secure data transmission.	- Integrate with Azure Virtual Network (VNet) and use Service Endpoints. - Configure network rules and firewalls. - Use private endpoints for secure access.
Monitoring and Logging	Monitor storage account activity and enable logging for audit and compliance.	- Use Azure Monitor for monitoring and alerting. - Enable Storage Analytics logging for audit trail and compliance checks.
Data Protection and Disaster Recovery	Ensure data availability and recovery mechanisms.	- Implement Geo-Redundant Storage (GRS) for data replication. - Use Azure Backup for critical data backup and recovery. - Establish disaster recovery plans.
Compliance and Governance	Adhere to regulatory requirements and industry standards.	- Regular security assessments and compliance audits. - Implement encryption and access controls per regulatory guidelines. - Maintain documentation and reporting.

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Authorization storage Account

- Authorization for Azure Storage accounts involves managing access permissions to storage resources to ensure secure and controlled operations.

Authorization Mechanism	Description	Use Cases	Best Practices
Azure Role-Based Access Control (RBAC)	Assign specific roles to users or groups at different levels of Azure resources.	- Control access to Azure Storage accounts, containers, and blobs. - Manage permissions based on job roles and responsibilities.	- Use built-in roles (e.g., Storage Account Contributor) or define custom roles with least privilege principles. - Regularly review and audit role assignments.
Shared Access Signatures (SAS)	Delegated access tokens with specific permissions and constraints.	- Provide temporary access to blobs, containers, or queues. - Enable fine-grained access control without exposing account keys.	- Generate SAS tokens with minimum necessary permissions and scope .- Set expiration times and IP restrictions to limit access duration and sources.
Azure Active Directory (Azure AD) Integration	Use Azure AD identities for authentication and access control.	- Manage access using Azure AD users, groups, and service principals. - Centralized identity management and leverage Azure AD features.	- Implement Azure AD authentication for secure access to Azure Storage resources. - Enable multi-factor authentication (MFA) and conditional access policies.
Service Principals and Managed Identities	Identities used by applications and services to authenticate to Azure resources.	- Grant permissions to applications and automation tasks accessing Azure Storage. - Enhance security by avoiding credential storage.	- Use managed identities for Azure resources (e.g., Azure VMs, Azure Functions) to automate access without credentials. - Limit permissions based on specific tasks and roles.

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- Considerations for Implementing Authorization:
    - Role Assignment: Assign roles and permissions based on organizational roles and responsibilities using Azure Portal, Azure CLI, or PowerShell.
    - SAS Token Management: Generate and manage SAS tokens with careful consideration of permissions, validity periods, and IP restrictions.
    - Azure AD Integration: Configure Azure AD authentication and leverage Azure AD features such as conditional access and identity protection.

Azure Data Factory (ADF)

- Azure Data Factory (ADF) is a cloud-based data integration service that allows you to create data-driven workflows for orchestrating and automating data movement and data transformation. It is designed to handle complex data pipelines and ETL (Extract, Transform, Load) processes, making it easier to integrate data from various sources.
- Key Features:
    - Data Movement:
        - Move data between various data stores, both on-premises and in the cloud, with high reliability and performance.
    - Data Transformation:
        - Perform data transformation at scale using data flows, mapping data flows, and external processing frameworks like HDInsight, Databricks, and Azure Machine Learning.
    - Orchestration and Scheduling:
        - Create complex data pipelines to orchestrate data movement and transformation, schedule jobs to run at specific times, and manage dependencies.
    - Integration:
        - Integrates seamlessly with a wide range of data sources, including Azure services (Azure Blob Storage, Azure SQL Database, Azure Data Lake), on-premises databases, and third-party services.
    - Monitoring and Management:
        - Provides comprehensive monitoring and management capabilities to track pipeline execution, diagnose issues, and ensure data processes run smoothly.
    - Code-Free and Code-Based Options:
        - Offers a visual interface for building data pipelines without writing code and supports code-based development using SDKs and APIs.
- Components of Azure Data Factory:
    - Pipelines: Logical grouping of activities that perform a unit of work.
    - Activities: Define the actions to be performed on data (e.g., data movement, data transformation, control flow).

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    - Datasets: Represent data structures within data stores, defining the schema and location of the data.
    - Linked Services: Define the connection information needed for Data Factory to connect to external data sources.
    - Triggers: Run pipelines on specified schedules, in response to events, or manually.
    - Integration Runtimes: Provide the compute infrastructure for data movement, data flow, and activity dispatch. Types include Azure, self-hosted, and Azure-SSIS integration runtimes.
- Security and Compliance:
    - Data Encryption: Data in transit and at rest is encrypted.
    - Access Control: Supports Azure Active Directory (AAD) integration and role-based access control (RBAC).
    - Compliance: Complies with industry standards and regulations like GDPR, HIPAA, and ISO..
- Integration and Extensibility:
    - Integration with Azure Services:
        - Seamlessly integrates with Azure services like Azure Synapse Analytics, Azure Databricks, and Azure Machine Learning.
    - Custom Activities:
        - Supports custom activities using Azure Batch or external compute environments.
    - Data Connectors:
        - Provides connectors to integrate with various data sources, including SQL databases, NoSQL databases, SaaS applications, and file systems.
- Monitoring and Management:
    - Monitoring:
        - Built-in monitoring dashboard to track the status of pipeline runs, activity runs, and integration runtimes.
    - Alerts and Notifications:
        - Supports setting up alerts and notifications for pipeline failures, successes, and other important events.
    - Logging:
        - Offers detailed logging capabilities to diagnose issues and troubleshoot pipeline execution.
- Azure Data Factory is a versatile and powerful tool for data integration and ETL processes, providing a comprehensive solution for orchestrating data workflows, transforming data, and integrating with various data sources and services both in the cloud and on-premises.

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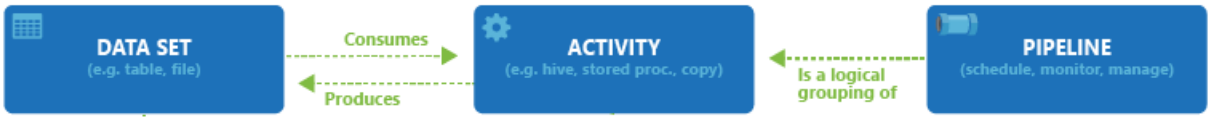
Pipelines & Activities

Pipelines

- A logical grouping of activities that perform a unit of work.
- Purpose: Organize and manage related data workflows.
- Key Features:
    - Logical Grouping: Manage and organize data workflows by grouping related activities.
    - Scheduling and Execution: Can be scheduled, triggered by events, or executed manually.
    - Error Handling and Retry Logic: Include error handling and retry logic for robustness.
    - Parameterization: Enable dynamic execution based on input parameters, making pipelines reusable.
- Example: A pipeline might extract data from a SQL database, transform it using a mapping data flow, and load the transformed data into Azure Data Lake.

Activities

- Actions performed on data within a pipeline. Each activity represents a single step in a process.
- Activities, also known as tasks or steps, are individual units of work within a pipeline that perform specific actions on data. Each activity typically represents a discrete data processing operation, such as data ingestion, transformation, validation, or analysis.
- In short, Activities in pipeline defines action to perform on data.



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Types of Activities

1. Data Movement Activities:
    - As the name suggests, these activities focus on transferring data between different sources and destinations. Common examples include copying data from databases, cloud storage services, or even on-premises systems.
    - E.g.
        - Azure Blob Storage
        - Azure Data Lake Storage
        - Common Data Service
        - SQL Server
        - Many more depending on specific connectors

2. Data Transformation Activities :
    - These activities manipulate and modify the data to suit your needs. This could involve cleaning the data, formatting it consistently, or performing more complex transformations using external compute services.
    - E.g.
        - Data cleansing tasks (removing duplicates, fixing errors)
        - Data formatting (changing data types, applying filters)
        - Using Azure Databricks or Azure HDInsight for complex transformations

3. Control Flow Activities:
    - These activities manage the execution flow of your pipeline. They allow you to control the sequence in which activities run, handle branching logic (like if-else statements), loop through data sets, or even wait for specific events.
    - E.g.
        - Set Variable - define and manage variables within the pipeline
        - Execute Pipeline - call and run another pipeline within the current one
        - For Each - iterate through a loop based on a data set
        - If Condition - perform actions based on conditions in the data
        - Wait and Webhook activities - handle delays or trigger external actions

- Pipelines: Group related activities into a logical unit of work. Can be scheduled, triggered, or executed manually.
- Activities: Individual steps within a pipeline that perform specific actions on data, such as data movement, transformation, and control flow.

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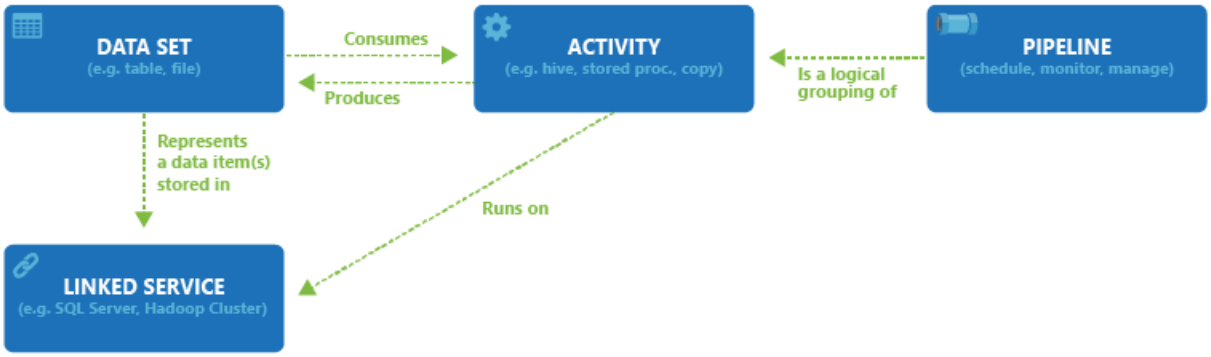
Linked Services and Datasets

Linked Services

- Linked Services are similar to connection strings in database systems. They define the connection information needed for Azure Data Factory to connect to external data sources.
- Purpose: Establish connections to data stores and compute services that activities in a pipeline can utilize.
- Key Features:
    - Connection Information: Stores details like authentication credentials, server names, and other necessary connection parameters.
    - Multiple Data Sources: Can connect to various types of data sources, including Azure services (e.g., Azure Blob Storage, Azure SQL Database), on-premises databases, SaaS applications, and third-party services.
    - Security: Securely manage sensitive information such as credentials and keys using Azure Key Vault integration.
- Example: A Linked Service could define a connection to an Azure SQL Database, including the server name, database name, and authentication details.

Data Sets

- Definition: Datasets represent data structures within data stores. They define the schema and location of the data to be used in activities.
- Purpose: Provide a structure for the data that activities will consume or produce, including the format and schema of the data.
- Key Features:
    - Data Schema: Define the structure and format of the data, such as columns in a database table or the structure of a JSON file.
    - Location: Specify the location of the data within the data store, such as a file path or table name.
    - Integration with Activities: Used by activities to read from or write to data stores, ensuring that data operations adhere to the defined schema and format.
- Example: A Dataset could represent a table in an Azure SQL Database, specifying the table name and the columns it contains, or it could represent a CSV file in Azure Blob Storage, specifying the file path and column structure.



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Aspect	Linked Services	Datasets
Definition	Connection configurations to data sources	Representation of data structure and metadata
Purpose	Establish connectivity to external data sources	Define how and where data is stored
Configuration	Contains connection details and settings	Defines schema, location, format, and properties of data
Reusability	Can be reused across multiple datasets/pipelines	Specific to individual datasets
Security	Supports secure authentication and encryption	Metadata representation, no data storage

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Debug & Triggers

Debug

- Process of testing and troubleshooting pipelines and activities before deployment.
- Purpose: Ensure pipelines and activities function correctly and meet requirements.
- Key Features:
  - Interactive Testing: Run pipelines interactively to observe and validate outputs.
  - Step-by-Step Execution: Execute activities sequentially for debugging purposes.
  - Real-Time Logs: View logs in real-time to track data flow and identify errors.
  - Parameter Testing: Test pipelines with different inputs to validate robustness.
- Example: Testing a pipeline that moves data from an on-premises database to Azure Data Lake to verify data integrity and transformation logic.

Trigger

- Mechanism to automate pipeline execution based on predefined conditions or schedules.
- Purpose: Automate data processing tasks without manual intervention.
- Types of Triggers:

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    - Schedule Trigger: Executes pipelines at specified intervals (e.g., hourly, daily).
    - Tumbling Window Trigger: Processes data in recurring, fixed-sized time windows.
    - Event-Based Trigger: Initiates pipeline runs based on events like file arrival or HTTP request.
- Key Features:
    - Automation: Ensures pipelines run according to defined schedules or events.
    - Flexibility: Supports various trigger types to accommodate different operational needs.
    - Dependency Management: Handles dependencies between pipelines and triggers for orderly execution.
    - Monitoring and Alerts: Provides monitoring dashboards and alerts for trigger status and performance.
- Example: Using a schedule trigger to run a pipeline daily at midnight for aggregating sales data from multiple sources into a data warehouse.

Trigger Type	Description	Configuration	Example
Schedule Triggers	Initiate pipeline execution at predefined time intervals	- Recurrence pattern (hourly, daily, weekly) - Start time, end time - Time zone - Days of the week - Interval frequency (e.g., every 1 hour)	Running a daily data ingestion pipeline
Event Triggers	Start pipeline execution in response to external events	- Event source (e.g., Azure Blob Storage, Azure Event Hubs) - Conditions - Actions to take when triggered	Triggering a pipeline on new file arrival
Tumbling Window Triggers	Execute pipelines based on predefined time windows	- Window size (e.g., hourly, daily) - Start time, end time - Recurrence pattern - Time zone	Aggregating hourly data into daily summaries
Manual Triggers	Allow manual initiation of pipeline execution	- User-initiated through UI, PowerShell, or REST API	Running a pipeline for ad-hoc data tasks

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Integration Runtime

- Integration Runtime (IR) in Azure Data Factory facilitates data integration by providing compute infrastructure to connect and move data across various environments securely.
- Integration Runtimes, the backbone of Azure Data Factory and Azure Synapse pipelines, are compute environments that facilitate data integration tasks like data movement and transformation across various network environments.
- Purpose:
    - Enables Azure Data Factory to:
    - Connect securely to on-premises data sources.
    - Move data between cloud and on-premises environments.
    - Execute data transformation activities efficiently.
- Types of Integration Runtimes:
    1. Azure Integration Runtime:
        - Hosted in Azure for data movement between Azure services.
        - Executes activities like copying data between Azure Blob Storage and Azure SQL Database.
    2. Self-Hosted Integration Runtime:
        - Installed on-premises or on a virtual machine (VM) within a private network.
        - Connects to on-premises data sources such as SQL Server or Oracle.
        - Performs data movement and transformation tasks locally.
    3. Azure-SSIS Integration Runtime:
        - Runs SQL Server Integration Services (SSIS) packages in Azure Data Factory.
        - Supports existing SSIS packages for data integration and transformation.
        - Utilizes Azure VMs for scalable SSIS package execution.
- Key Features:
    - Security: Ensures data security with network isolation and encryption.
    - Scalability: Scales compute resources based on workload demands.
    - Versatility: Supports connectivity to diverse data sources, cloud or on-premises.
- Examples:
    - Using Self-Hosted Integration Runtime to connect Azure Data Factory to on-premises SQL Server.
    - Deploying Azure-SSIS Integration Runtime for executing SSIS packages in Azure, integrating data across systems.

Type of Integration Runtime	**Azure Integration Runtime**	**Self-Hosted Integration Runtime**	**Azure-SSIS Integration Runtime**
Description	Hosted in Azure, facilitates data	Installed on-premises or on a virtual machine (VM) in a	Specifically designed for running SQL Server Integration

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	movement between Azure services and cloud-based data stores.	private network, connects to on-premises data sources and performs data movement locally.	Services (SSIS) packages in Azure Data Factory, it supports execution of SSIS packages in Azure VMs.
Purpose	Connect securely to Azure data services and perform data movement within the cloud.	Enable connectivity to on-premises data sources like SQL Server, Oracle, and files stored locally.	Execute existing SSIS packages in the cloud for data integration and transformation tasks.
Key Features	- Secure data movement - Scalable compute resources - Integration with Azure services	- On-premises connectivity - Local execution of data movement and transformation tasks - Secure communication with Azure Data Factory	- Compatibility with existing SSIS packages - Scalable execution with Azure VMs - Integration with Azure Data Factory for comprehensive data workflows
Examples	Copying data between Azure Blob Storage and Azure SQL Database.	Connecting Azure Data Factory to an on-premises SQL Server database for data extraction and loading.	Running SSIS packages in Azure to integrate data from various sources into a centralized data warehouse.

Pipeline Parameters and Variables

A. Pipeline Parameters:
   - Placeholders to pass values into pipelines at runtime.
   - Purpose: Enable dynamic execution and reusability of pipelines.
   - Key Features:
     - Data types and default values.
     - Value passing from triggers or other pipelines.
   - Example: Defining a folder path in Azure Blob Storage based on different input scenarios.

B. Variables:
   - Hold and manipulate values within pipeline executions.
   - Purpose: Support data transformation and control flow in pipelines.
   - Key Features:
     - Scope (global or local) and data types.
     - Use of expressions for dynamic value assignment.
   - Example: Storing current date for file naming or calculation purposes.

Component	**Pipeline Parameters**	**Variables**
Description	Placeholders for passing values into pipelines at runtime.	Hold and manipulate values within pipeline executions.

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Purpose	Enable dynamic execution and reusability of pipelines.	Support data transformation and control flow in pipelines.
Key Features	- Data types and default values - Value passing from triggers or other pipelines.	- Scope (global or local) and data types - Use of expressions for dynamic value assignment.
Example	Defining a folder path in Azure Blob Storage based on different input scenarios.	Storing current date for file naming or calculation purposes.

System Variables

- System Variables in Azure Data Factory are predefined variables that provide information about the runtime environment and execution context of pipelines and activities.
- Purpose:
  - System Variables serve various purposes, including:
  - Monitoring: Tracking execution progress and status.
  - Control Flow: Controlling the behavior and flow of pipeline activities.
  - Dynamic Behavior: Supporting dynamic configuration and decision-making during runtime.
- Key Features:
  - Read-Only: System Variables are read-only and cannot be modified during pipeline execution.
  - Dynamic Access: Accessed using expressions within activities and pipelines to dynamically adjust behavior based on runtime conditions.
  - Visibility: Available across various components of Azure Data Factory, enhancing operational visibility and control.
- Example Use:
  - Using Pipeline().RunId to log execution details and traceability for audit purposes.
  - Employing Activity().Name to conditionally branch workflows based on the current activity's status or duration.

Category	Variable	Description	Example
Pipeline Execution Variables	Pipeline().PipelineName	Retrieves the name of the current pipeline being executed.	Using Pipeline().PipelineName to log pipeline execution details.
	Pipeline().RunId	Provides a unique identifier for the current pipeline run instance.	Using Pipeline().RunId for traceability and auditing of pipeline executions.
	Pipeline().TriggerType	Indicates the type of	Checking

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		trigger (e.g., schedule, event) that initiated the pipeline run.	Pipeline().TriggerType to handle different trigger types dynamically.
Activity Execution Variables	Activity().Name	Retrieves the name of the current activity being executed within a pipeline.	Using Activity().Name to conditionally branch workflows based on activity status.
	Activity().StartTime	Provides the start time of the current activity run.	Logging Activity().StartTime to track activity execution times.
	Activity().Duration	Indicates the duration of the current activity run.	Monitoring Activity().Duration to measure performance of activities.
Data Flow Activity Variables	DataFlow().IsDebugEnabled	Indicates whether debug mode is enabled for a data flow activity.	Using DataFlow().IsDebugEnabled to conditionally enable debug behavior in data flows.
	DataFlow().RunId	Provides a unique identifier for the current data flow activity run.	Using DataFlow().RunId for traceability and debugging within data flows.
Trigger Execution Variables	Trigger().TriggerType	Specifies the type of trigger that initiated the pipeline execution.	Checking Trigger().TriggerType to dynamically adjust pipeline behavior based on trigger type.
	Trigger().ScheduledTime	Provides the scheduled time at which the trigger initiated the pipeline run.	Using Trigger().ScheduledTime to log scheduled execution times for reporting purposes.
	Trigger().Time	Provides the current timestamp when the trigger fires.	Logging Trigger().Time to track precise trigger execution times.

Connectors in Azure Data Factory (ADF)

- Connectors in Azure Data Factory are integration interfaces that enable seamless data movement and transformation between various data sources and destinations.
- Purpose:
    - Data Integration: Enable seamless movement of data between diverse data sources and destinations.
    - Transformation: Support data transformation activities such as filtering, aggregating, and mapping data.
    - Interoperability: Facilitate integration across hybrid environments, connecting cloud-based and on-premises data stores.

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- Key Features:
    - Versatility: Support for various data types and formats, ensuring compatibility across different systems.
    - Scalability: Ability to handle large volumes of data efficiently, crucial for big data and analytics workflows.
    - Security: Ensure secure data transmission and compliance with data governance policies.
- Usage Examples:
    - Azure Blob Storage Connector: Copying files from on-premises servers to Azure Blob Storage.
    - SQL Server Connector: Extracting data from an on-premises SQL Server database for transformation and loading into Azure SQL Database.

Type of Connector	Description	Examples
Azure Connectors	Integration interfaces for Azure services facilitating data movement within Azure.	Azure Blob Storage, Azure SQL Database, Azure Data Lake Storage, Azure Cosmos DB, Azure Synapse Analytics
External Connectors	Integration interfaces for external data sources and platforms, enabling hybrid data integration.	SQL Server, MySQL, PostgreSQL, Oracle Database, Salesforce, Google BigQuery, Amazon S3

Difference Between Connectors and Linked Services

Aspect	Connectors	Linked Services
Definition	Integration interfaces enabling data movement and transformation	Configuration entities defining connection details
Purpose	Facilitate data interaction between data stores and ADF pipelines	Store connection information and authentication details
Functionality	Provide technical interface for data interaction	Enable secure connections and data access
Examples	Azure Blob Storage Connector, SQL Server Connector	Azure Blob Storage Linked Service, SQL Server Linked Service
Usage in ADF	Defined within activities and data flows for data movement	Configured at the dataset level to establish data connections

- Summary:
    A. **Connectors** are technical interfaces that define how Azure Data Factory interacts with specific data sources or services.
    B. **Linked Services** are configuration entities within Azure Data Factory that store connection information required by connectors to establish connections with external data sources or computing services.

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ADF Activities

Activity Type	Description
Copy Data Activity	Copies data from a source to a destination data store, including operations like serialization/deserialization, compression/decompression, and column mapping.
Delete Activity	Deletes files or folders from on-premises or cloud storage, useful for cleaning up or archiving files when they are no longer needed.
Set Variable Activity	Defines and assigns values to variables within a pipeline. Variables can be of type String, Bool, or Array.
Append Variable Activity	Appends values to an existing array variable within a pipeline. Can only be used with variables of the Array type.
Execute Pipeline Activity	Invokes another pipeline from within a pipeline, allowing for the creation of modular pipelines where one master pipeline calls multiple child pipelines.
Filter Activity	Selects specific data records based on conditions, applying a filter expression to an input array.
ForEach Activity	Iterates over a collection, executing specified activities for each item, similar to a ForEach loop in programming.
Get Metadata Activity	Retrieves metadata about data entities within ADF pipelines, such as file size, creation date, last modified date, and more.
If Condition Activity	Executes a set of activities based on the evaluation of a condition, similar to an if statement in programming.
Wait Activity	Pauses pipeline execution for a specified duration before continuing with subsequent activities.
Until Activity	Repeatedly executes a set of activities until a specified condition evaluates to true, functioning like a do-until loop in programming.
Web Activity	Calls custom REST endpoints from within a pipeline, making HTTP requests to web services or APIs.
Switch Activity	Executes a set of activities based on evaluated expressions, similar to a switch statement in programming.
Validation Activity	Verifies that a pipeline proceeds only after confirming that the linked dataset exists or meets specified criteria.
Lookup Activity	Fetches data from a designated data source for use within pipelines, allowing dynamic determination of objects for subsequent activities.
Data Flow Activity	Executes visually designed data transformation processes (Mapping Data Flows) within ADF pipelines, utilizing scaled-out Apache Spark clusters.

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Azure Key Vault

- Azure Key Vault is a cloud service provided by Microsoft Azure that helps safeguard cryptographic keys and secrets used by cloud applications and services.
- Purpose:
    - Security: Ensures secure storage of sensitive information such as API keys, passwords, certificates, and cryptographic keys.
    - Management: Simplifies the management of secrets, keys, and certificates, enabling centralized control and monitoring.
    - Access Control: Allows fine-grained access control to secrets and keys, ensuring that only authorized applications and users can access them.
- Key Features:
    1. Secrets Management:
        - Store and tightly control access to tokens, passwords, certificates, API keys, and other secrets.
        - Example: Storing database connection strings securely.
    2. Key Management:
        - Create, import, and manage cryptographic keys used for encryption and decryption.
        - Example: Generating and managing encryption keys for data protection.
    3. Certificate Management:
        - Provision, manage, and deploy SSL/TLS certificates for secure communications.
        - Example: Automating certificate renewal and deployment for web applications.
    4. Access Policies:
        - Define access policies to grant and restrict access to secrets and keys based on roles and permissions.
        - Example: Configuring policies to allow only specific applications to access particular secrets.
    5. Integration with Azure Services:
        - Seamless integration with other Azure services like Azure Active Directory, Azure Storage, and Azure App Service.
        - Example: Using Key Vault to manage secrets for an Azure App Service application.
    6. Logging and Monitoring:
        - Enable logging to monitor access and usage of keys and secrets, enhancing security and compliance.
        - Example: Tracking access attempts and changes to secrets for auditing purposes.
- Usage Examples:
    - Secure Configuration Management: Store and manage application configuration settings securely.
    - Encryption Key Management: Manage keys for encrypting data at rest and in transit.

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    - Certificate Management: Automate the lifecycle of SSL/TLS certificates.
- Azure Key Vault helps organizations enhance security, simplify key and secret management, and ensure compliance with industry standards and regulations.

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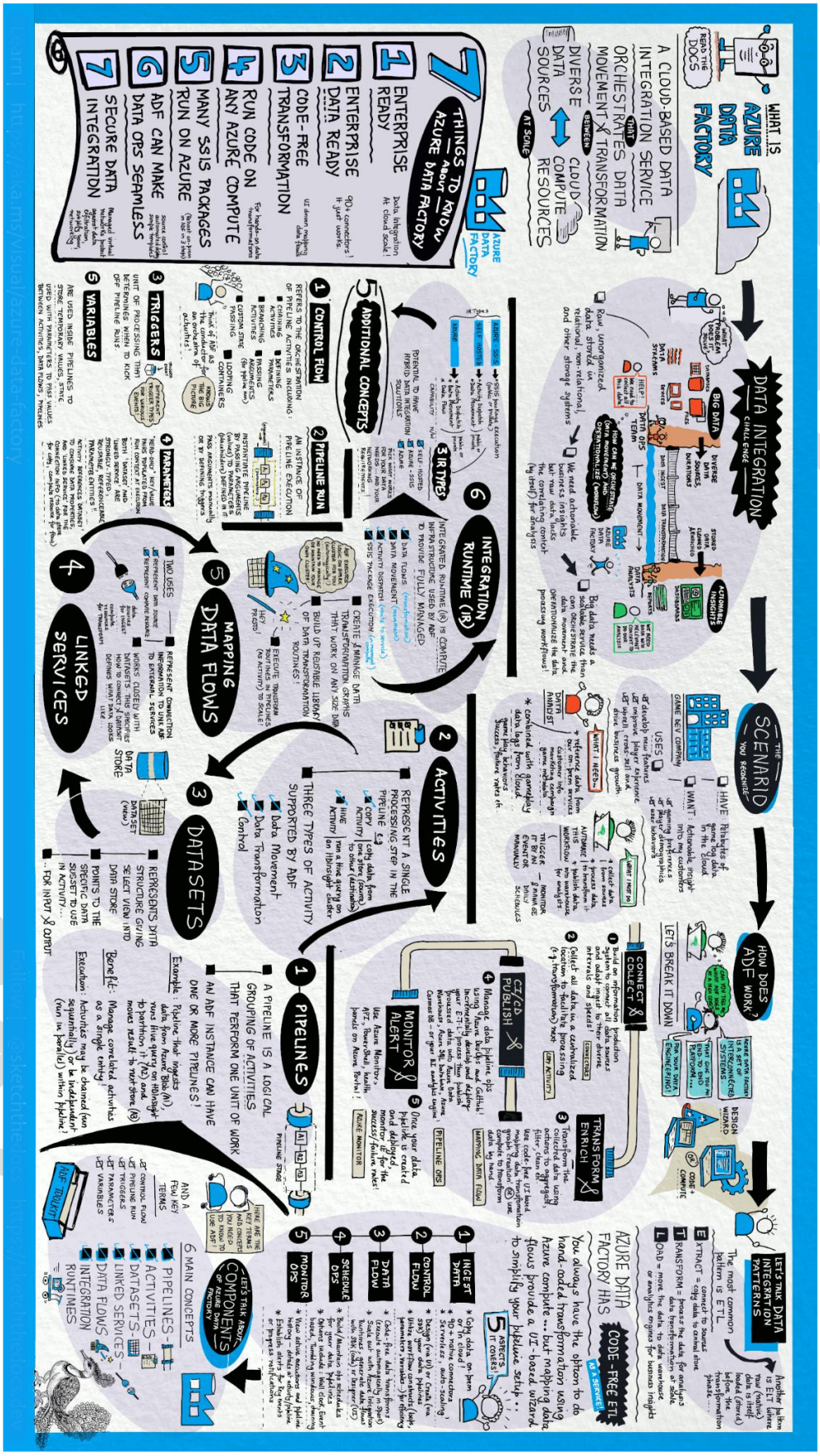
Azure Data Factory End-to-End Concept Map